Answer key: Lesson 3, Copy on the Grid

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Mei draws an L shape on a grid.

Answer: A copy of the L shape; 3; 3

The L shape is 3 steps across and 3 steps up. To copy it, start at the top left dot of the empty grid and draw: 1 right, 2 down, 2 right, 1 down, 3 left and 3 up. The copy has the same six corners in the same places on its own grid.

Watch for: Counting dots, not steps (4 and 4).; Making the bottom bar too short or too long by one step.

S2. A counter starts at S. It moves only right and up to reach E.

Answer: 2 right, 3 up; any path that goes only right and up

E is 2 steps to the right of S and 3 steps up. One path is 2 right and then 3 up. Any path with 2 right steps and 3 up steps works, in any order.

Watch for: Counting dots, not steps (3 right).; Swapping right and up (3 right, 2 up).

S3. Start at S. Go 4 steps right. Then go 1 step up.

Answer: A line 4 right and then 1 up; ring on the dot where it stops

Start on S. 4 steps right: count 1, 2, 3, 4 jumps along the bottom row. Then 1 step up. The ring goes on the dot 4 right and 1 up from S.

Watch for: Counting S as the first step.; Going 4 up and 1 right (swapped).

S4. Which picture is an exact copy of the T shape?

Answer: B

The T shape has a top bar 3 steps across and a stem that makes it 3 steps tall. Picture A has a stem 1 step too short. Picture C has a top bar 1 step too wide. Picture B is the same T shape moved 1 step to the right, so B is the exact copy.

Watch for: Choosing the picture that sits in the same place as the T (a copy can be in a different place).; Counting dots instead of steps.

S5. Three corners of a rectangle are ringed. Join the dots to finish the rectangle.

Answer: 3 right, 2 up

One ringed corner is 3 steps right of S. The other is 2 steps up from S. The missing corner has both moves: 3 right and 2 up. Join the four corners to make the rectangle.

Watch for: Putting the corner in the same row or column as S.; Swapping right and up (2 right, 3 up).

S6. Look at the path from S to E. Tick True or False for each sentence.

Answer: True; False; False; True

The path goes 5 steps right and 1 step up. The bottom line touches 6 dots, because a line of 5 steps has 6 dots. So: 5 steps right is true; 2 steps up is false (1); 5 dots is false (6); E is 5 right and 1 up is true.

Watch for: Ticking True for the number of dots.; Counting the starting dot as a step.

Think harder (page 3)

T1. A counter goes from S to E. It moves only right or up. The grid is 3 steps across and 1 step up.

Answer: U R R R, R U R R and R R U R (any order); 4 routes

Every route has 3 right steps and 1 up step. The U can be first, second, third or last: U R R R, R U R R, R R U R and R R R U. That is 4 routes.

Watch for: Writing the same route twice.; Stopping at 3 routes and missing one.

T2. A counter goes from S to E. It moves only right or up. The grid is 2 steps across and 2 steps up.

Answer: R R U U, R U R U and R U U R (any order); 6 routes

Routes that start with R: R R U U, R U R U and R U U R. Routes that start with U are U U R R, U R U R and U R R U. 3 and 3 make 6 routes. Each route has 2 R and 2 U; the two R can sit in 6 pairs of places.

Watch for: Writing a route that starts with U again.; Counting 4 routes.

T3. Ravi's counter walks 3 right and 1 up, again and again. The picture shows 2 sets.

Answer: 9 right, 3 up; 15 right, 5 up

One set is 3 right and 1 up. 2 sets: 6 right and 2 up. 3 sets: 9 right and 3 up. 5 sets: 3, 6, 9, 12, 15 right and 5 up.

Watch for: Adding 3 right only once.; Forgetting the steps up.

T4. Start at S. Go 3 steps right. Go 2 steps up. Then go 1 step left.

Answer: 2 right, 2 up

3 right then 1 left leaves 2 right of S. The 2 up stay. You stop 2 right and 2 up from S.

Watch for: Writing 3 right (forgetting the left step).; Moving 1 step right instead of left.

Go further (page 4)

G1. The boxes show how many routes reach each corner. S is the bottom left box. E is the top right box.

Answer: 4 (middle row); 3, 6 and 10 (top row); 10 routes

Each box is the box on its left plus the box below it. Middle row: 3 + 1 = 4. Top row: 1 + 2 = 3, then 3 + 3 = 6, then 6 + 4 = 10. So 10 routes reach E on a grid 3 across and 2 up.

Watch for: Writing 1 in every empty box.; Adding the box above instead of the box below.

G2. A counter goes from S to E, moving only right or up. The grid is 3 steps across and 2 steps up. The route must pass the ringed corner.

Answer: 3; 2; 6

To the ringed corner: 2 right and 1 up gives 3 routes (R R U, R U R, U R R). From it to E: 1 right and 1 up gives 2 routes (R U, U R). Each of the 3 first parts can be followed by either of the 2 second parts: 3 groups of 2 is 6 routes.

Watch for: Adding 3 and 2 to get 5.; Counting all 10 routes on the grid.

G3. Draw a shape with 4 corners on the grid. A is its bottom left corner. Many answers are right.

Answer: Many answers are right: three more corners, each said as moves right and up from A, that join A to make a shape

For example, a rectangle: B is 3 right and 0 up, C is 3 right and 2 up, D is 0 right and 2 up. A friend can copy the shape by marking each corner from A and joining the corners.

Watch for: Counting dots, not steps.; Giving a corner that is left of A.

Investigation (page 5): Same steps, different routes

Both routes have 5 steps. Every route from S to E needs 3 steps right and 2 steps up, so every route has 5 steps. Only the order of the steps changes. (There are 10 routes on this grid.)

Look for: A child who says every route needs the same moves: 3 right and 2 up, only in a different order. Credit any clear reason. A child who draws two routes and counts 5 and 5 has the first step.

Exit ticket (page 5)

E1. A counter starts at S and moves only right and up to reach E.

Answer: 2 right, 2 up

E is 2 steps right of S and 2 steps up. Count the jumps, not the dots.

Watch for: Counting dots (3 and 3).; Swapping right and up.

E2. A counter moves 2 right and 3 up. Then it moves 3 right and 1 up.

Answer: 5 right, 4 up

2 and 3 make 5 right. 3 and 1 make 4 up.

Watch for: Adding the rights and the ups together (9).; Writing 3 right and 1 up (the last move only).

E3. A figure is 5 steps across and 2 steps up. A child draws a copy that is 6 steps across and 2 steps up.

Answer: No; 5

An exact copy is the same size. 6 steps across is 1 step too wide, so it is not an exact copy. The copy should be 5 steps across.

Watch for: Saying Yes because the height is the same.; Writing 6 for part b.

Sneak-peek puzzles (not in the workbook)

1. 3 right, 1 up.
2. 2 routes: right then up, or up then right.